

Quiz (Carolina Reaper)

- Q1.** Factor the expression $x^3 + 8$
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 + 2x + 4)$
(D) $(x - 2)(x^2 - 2x + 4)$
(E) Cannot be factored
- Q2.** Factor the expression $27x^3 - 8y^3$
(A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(C) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
(D) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
(E) Cannot be factored
- Q3.** Which of the following is a factor of $x^3 + 64$?
(A) $x + 4$
(B) $x - 4$
(C) $x + 2$
(D) $x - 2$
(E) $x + 8$
- Q4.** Factor the expression $8x^3 - 1$
(A) $(2x - 1)(4x^2 + 2x + 1)$
(B) $(2x + 1)(4x^2 - 2x + 1)$
(C) $(2x - 1)(4x^2 - 2x + 1)$
(D) $(2x + 1)(4x^2 + 2x + 1)$
(E) Cannot be factored
- Q5.** Which of the following is equivalent to $(x + 2)(x^2 - 2x + 4)$?
(A) $x^3 + 8$
(B) $x^3 - 8$
(C) $x^3 + 2$
(D) $x^3 - 2$
(E) $x^2 + 2$
- Q6.** Factor the expression $x^3 - 27$
(A) $(x - 3)(x^2 + 3x + 9)$
(B) $(x + 3)(x^2 - 3x + 9)$
(C) $(x - 3)(x^2 - 3x + 9)$
(D) $(x + 3)(x^2 + 3x + 9)$
(E) Cannot be factored
- Q7.** Which of the following is a factor of $64 - x^3$?
(A) $x - 4$
(B) $x + 4$
(C) $x - 2$
(D) $x + 2$
(E) $x + 8$
- Q8.** Factor the expression $27x^3 + 8y^3$
(A) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(B) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(C) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
(D) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
(E) Cannot be factored

Open Ended Questions (FRQ)

- Q9.** Factor the expression $x^3 - 125$ and explain your reasoning.
- Q10.** Factor the expression $8x^3 + 27$ and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to check for common factors before factoring.
- Tip 2: Use the sum and difference of cubes formulas to factor expressions.
- Tip 3: Check your work by multiplying the factors to ensure you have factored correctly.